



# UrbanHive

Facility Technologies Pvt. Ltd.

Ops(TM) | Apartment Maintenance Intelligence System

CONFIDENTIAL  
Contractor & Vendor Use Only  
Bengaluru, India

## TECHNICAL MAINTENANCE MANUAL

# Water Pump Maintenance Manual - UH-PUMP-405

|                |                                                     |
|----------------|-----------------------------------------------------|
| Document ID    | MAN-030                                             |
| Version        | v1.0.0                                              |
| Client         | UrbanHive Facility Technologies Pvt. Ltd.           |
| Site           | GreenPark Meridian Residency, Whitefield, Bengaluru |
| Effective date | 2024-07-01                                          |
| Review cycle   | Annual or after any major incident                  |

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## UrbanHive Ops - Contractor Use

support: [maintenance@urbanhiveops.example.com](mailto:maintenance@urbanhiveops.example.com) | AMIS helpdesk ext. 4400

## 1. Equipment Overview and Technical Specifications

End-suction centrifugal booster pump for domestic water header pressure maintenance at GreenPark Meridian. Stainless impeller, mechanical seal, pressure transducer, and VFD control maintaining 2.8-3.2 bar setpoint.

Model code: UH-PUMP-405

Asset registry: Record serial number and installation tower in AMIS before first energization.

Electrical supply: 415V 3-phase 7.5 kW, IP55 motor, VFD with pressure PID loop

Environmental limits: Ambient 0-45°C operational; store spare parts below 35°C and 60% RH.

Standards: UrbanHive engineering bulletin UH-ENG-AMIS-2024; applicable Indian Standards as cited per asset class.

Documentation package: This manual, wiring diagram sheet WD-UH-PUMP-405, commissioning checklist CC-030, and spare parts catalogue SP-UH-PUMP-405.

Technicians must verify nameplate data matches AMIS asset record UH-PUMP-405 before applying procedures in this document. Discrepancies must be logged as a data quality ticket to UrbanHive Data Engineering within one business day.

## 2. Installation Guidelines

Install on inertia base with flex connectors on suction and discharge. Prime pump before first start. Calibrate pressure transducer 0-6 bar against dead-weight tester. Register curve in AMIS.

### Pre-commissioning checks:

- Verify incoming power quality: voltage balance within 2%, THD < 5% on dedicated feeder.
- Confirm earthing resistance  $\leq 1$  ohm at equipment location; bond to society main earth bar.
- Inspect shipping damage; photograph defects for vendor warranty within 48 hours.
- Complete mechanical alignment, torque logs, and insulation resistance tests per UrbanHive QC form UH-QC-INST-02.

Post-installation: Upload signed commissioning certificate, as-built photos, and warranty start date to AMIS. Tag physical asset with QR code linking to this manual revision v1.0.0.

## 3. Preventive Maintenance Schedule

UrbanHive mandates the following intervals for GreenPark Meridian. Adjust only with written approval from Head of Maintenance.

- Daily: suction strainer ?P log, abnormal noise check.
- Weekly: gland leakage inspection, motor bearing temperature IR scan.
- Monthly: coupling alignment check, seal flush flow verification.
- Quarterly: vibration analysis; impeller wear visual via drain port.
- Annual: mechanical seal replacement; bearing grease per SKF chart.

Documentation: Each PM visit must generate a service log in AMIS with parts consumed, meter readings, and pass/fail checklist. Missed PM windows trigger automatic escalation to Facility Manager after 72 hours.

## 4. Troubleshooting Guide

Use isolation procedure: observe symptom ? confirm with meter/test ? apply remediation ? functional test ? document in AMIS.

If fault persists after two visits, escalate to UrbanHive vendor support with fault code and photos.

#### **Fault 01 - Code P101**

Symptom: Failure to build pressure

Remediation: Verify suction valve open; prime casing; check for air lock in rising main.

#### **Fault 02 - Code P203**

Symptom: Cavitation noise at impeller

Remediation: Increase suction head; clean strainer; reduce flow below NPSH available.

#### **Fault 03 - Code P310**

Symptom: Mechanical seal leak > drip rate

Remediation: Adjust seal flush; replace seal faces if scoring visible.

#### **Fault 04 - Code P405**

Symptom: VFD overcurrent trip

Remediation: Measure motor current imbalance; inspect impeller for obstruction.

#### **Fault 05 - Code P512**

Symptom: Pressure hunting  $\pm 0.4$  bar

Remediation: Retune PID gains; verify transducer snubber not clogged.

#### **Fault 06 - Code P601**

Symptom: Motor bearing over-temperature

Remediation: Grease per interval; check alignment; replace bearing if play > 0.05mm.

#### **Fault 07 - Code P708**

Symptom: Dry run protection false trip

Remediation: Clean electrodes in sump; verify cable shield grounding.

#### **Fault 08 - Code P802**

Symptom: Coupling rubber spider cracked

Remediation: Replace spider element; realign within 0.1mm/100mm.

#### **Fault 09 - Code P901**

Symptom: Check valve chatter

Remediation: Inspect disc seat for debris; verify spring tension spec.

#### **Fault 10 - Code P1004**

Symptom: Unusual vibration at 2 $\times$  line frequency

Remediation: Tighten base bolts; inspect foundation grout cracks.

#### **Fault 11 - Code P1102**

Symptom: Header tank overflow linked event

Remediation: Test float switch continuity; verify stop setpoint in PLC.

Fault 12 - Code P1203

Symptom: Discharge pressure below SLA 2.5 bar

Remediation: Parallel standby pump test; inspect non-return valve stuck closed.

5. Spare Parts List

Maintain minimum stock at GreenPark Meridian central store (Basement B1, Room ST-04).

| Part No. | Description                 | Min stock |
|----------|-----------------------------|-----------|
| SP-P01   | Mechanical seal kit         | 4         |
| SP-P02   | Coupling spider             | 6         |
| SP-P03   | Pressure transducer 0-6 bar | 2         |
| SP-P04   | Strainer basket SS316       | 3         |
| SP-P05   | VFD cooling fan             | 2         |
| SP-P06   | Gland packing set           | 5         |

Reorder when stock falls below minimum. Use UrbanHive procurement portal PR-UH-OPS for approved vendors only.

6. Safety Warnings and Compliance

Isolate electrical and lock-out before seal work. Drain line pressure before disassembly. Confined space rules apply for basement pump rooms. UrbanHive permit required for welding on discharge header.

Permit requirements: Hot work, confined space, work-at-height, and electrical isolation require digital permit in AMIS before start.

PPE minimum: Safety shoes IS 15298, helmet IS 2925, eye protection IS 5983, gloves appropriate to task.

Incident reporting: Any injury, near-miss, or property damage - complete IR-01 within 30 minutes; notify security desk ext. 9911.

7. Manufacturer Contact and UrbanHive Support

OEM technical support: +91-80-5555-0030 (08:00-20:00 IST)

Email: support-uh-pump-405@urbanhive-vendor.example.com

UrbanHive Ops escalation: maintenance@urbanhiveops.example.com - Priority 1 response within 4 business hours

AMIS helpdesk: Log ticket category `Water Pump` with asset\_id from tower registry

Emergency (life safety assets): 24x7 hotline +91-80-5555-0911

Revision history: v1.0.0 - Initial release for GreenPark Meridian AMIS integration and UrbanHive contractor certification programme.

End of document MAN-030